Title Name: Graph-Based Analysis of Movie Preferences

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1. Introduction:

This project addresses the problem of understanding user movie preferences through a graph-based approach. We aim to analyze the MovieLens dataset by modeling it as an undirected bipartite graph, where nodes represent users and movies, and edges represent ratings. By projecting this bipartite graph onto user-user and movie-movie relationships, we can identify clusters of users with similar preferences and movies that are frequently liked together. This study will help in refining movie recommendation systems and understanding influential users in shaping recommendations.

The need for this project arises due to the increasing demand for personalized movie recommendations and the challenge of effectively analyzing large-scale rating data. Our approach will provide insights into user clustering, genre-based movie similarities, and the identification of key influential users in the recommendation process.

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2. Objectives:

The main objectives of this project are:

- To analyze user clustering based on movie preferences using graph projections.
- To implement similarity-based movie recommendations by filtering and projecting movies.
- To evaluate the influence of users on recommendations using PageRank and edge centrality.

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3. Methodology:

The project will be carried out in the following steps:

1. **Data Collection:**
    - Utilize the MovieLens dataset (ml-latest-small), which contains 100,836 ratings from 610 users for 9,742 movies.
    - MovieLens Dataset: https://grouplens.org/datasets/movielens/
    - Extract relevant data from links.csv, ratings.csv, movies.csv, and tags.csv

2. **Analysis & Implementation:**
    - Construct an undirected bipartite graph with users and movies as nodes and ratings as edges.
    - **User Projection:** Convert the bipartite graph into a user-user graph where edge weights are based on cosine similarity. Apply a threshold to prune weak connections and identify clusters of users with similar movie preferences.
    - **Movie Projection:** Remove all edges with ratings less than 3. Construct a movie-movie graph where edge weights represent the number of common users who rated both movies. Apply a threshold to prune weak connections and identify movie clusters for recommendation.
    - **Genre Analysis:** Evaluate whether movies in a cluster belong to the same genres and assess the effectiveness of similarity-based recommendations.
    - **Influential Users Identification:**
        - Use PageRank to identify users who significantly influence recommendations.
        - Compute edge centrality to identify users with diverse movie ratings across genres (hubs in the network).
3. **Evaluation:**
    - Assess user clustering using centrality measures like degree centrality and betweenness centrality to identify key users in the network.
    - Evaluate movie recommendations using precision and recall by comparing recommended movies with user preferences.
    - Identify influential users using PageRank and eigenvector centrality to determine their impact on recommendations.

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4. Expected Outcomes:

- Identification of user clusters with similar movie preferences, aiding in personalized recommendations.
- Development of a movie similarity graph that improves recommendation accuracy.
- Insights into genre-based clustering of movies to validate recommendations.
- Identification of influential users who have a significant impact on the recommendation system.
- Contribution to research in graph-based recommendation systems, with potential applications in streaming services and online movie platforms.
